

David Xchel Morales Hurtado

Software Engineer — Automation, Validation & Infrastructure Tooling

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Professional Summary

Software Engineer with 5+ years of professional experience building automation, validation and infrastructure tooling on **Linux** platforms. Strong background in **C++**, **Python**, automation and large scale test frameworks. Experienced in **developing** reliable engineering solutions, **mentoring** and **collaborating** across **global organizations**. Passionate about **Open-Source** Software, **Mathematics** and continuous learning. Always aiming towards high **Quality** and **performance** with a **mentoring mindset** in favor of the continuous growth of the team.

Selected Achievements

- **Bronze Medalist** - Mexican Mathematics Olympiad (OMM).
- **Reduced validation execution time** from **hours to minutes** by building **Python/Bash automation frameworks**.
- **Improved testing efficiency by 50%** across multiple processor validation programs.
- **Served as technical mentor and onboarding lead** for incoming engineers.
- **Active open-source contributor**: Fixed a **custom property type bug** in **Blender**, resolved an **overflow bug** in the **fmt** library, removed a **duplicate-buffer deletion** in the **Godot Engine**, added a **large graph cycle detection test** to **CXXGraph**, and extended the test base of the **UnifyAI** Python project.
- Completed an **international academic exchange** at **Universiti Malaya**.

Professional Experience

Oracle — Guadalajara, Mexico
Senior Software Reliability Developer

Feb 2026 – Present

Develop automation and **cloud-management** tooling in **Python**, **Java**, and **Bash** for Oracle Cloud Infrastructure regions.

- Build and maintain automation workflows aimed at reducing manual effort, improving **reliability**, and **accelerating** cloud-region deployment activities.
- Troubleshoot **Linux** systems and **cloud** infrastructure issues across production environments.
- **Develop AI-assisted tooling** for information extraction, documentation, and operational support.
- **Authored** technical workflows and onboarding material adopted across engineering teams.

Intel — Guadalajara, Mexico
Functional Validation Engineer

Feb 2021 – Feb 2026

Develop **Python** and **Bash** automation for the functional validation of next-generation processor architectures on **Linux**, collaborating with a **multicultural** and **diverse** globally distributed team.

- Designed and maintained large-scale validation automation frameworks using **Python**, **Bash**, and **PyTest**.
- **Reduced execution time** of manual workflows from hours to minutes through **automation**.
- **Built** migration tooling that reduced project-porting effort **from one day to minutes**.
- **Developed CI/CD** validation pipelines supporting multiple processor programs.
- Acted as **technical lead** and primary **point of contact** across cross-functional teams.
- **Mentored** new engineers on processor architecture, memory systems, and **Linux**-based development workflows.
- Improved validation **efficiency by 50%** through **optimized** test coverage and execution strategy.

Education

Instituto Politécnico Nacional — Mexico City
B.Eng. in Mechatronics Engineering

Aug 2015 – Jan 2021

Top-performing student recognized for **effort**, **curiosity**, and **creativity**. Consistently focused on **teamwork** by **mediating** and **leading** to deliver quality projects efficiently.

- **MIPS** processor implementation in **SystemVerilog** with a **UVM testbench** built using **Python** and **C++**.
- **Python** projects: **Neural Networks**, **file/database processing**, **web scraping**, **Flask** applications with Selenium testing.
- **Virtual reality** experiences developed in **Unity** and **Blender**.
- Embedded systems and FPGA development projects.
- Electronic circuit design and implementation.

Academic Mobility / Exchange Program

Learned from one of the most renowned universities in the world while acquiring experience by working, **communicating** and **leading** people with contrasting backgrounds and mindsets.

- Built a line-following robot using **Raspberry Pi**.
- Interactive **VR** world using **Unity**.
- Developed a hand-tracking **AI** project.

Skills

Languages	Python, C++, Bash, C, Java, JavaScript, TypeScript, LaTeX, Verilog, Assembly
Platforms	Linux (Ubuntu, Debian, Arch), Git/GitHub, Docker
Engineering	Agile, CI/CD, Automation, Test Frameworks, Mathematics, AI-Assisted Development, Embedded Systems, Cloud Infrastructure
Professional	Leadership, Mentoring, Technical Communication, Problem Solving
Spoken Languages	English (fluent), Spanish (native), German (learning), French (learning)

Personal Projects

- **Modular C++ GUI application** (GTK4, webview, asio, GTest) — Continuously evolving desktop app featuring an embedded browser, a multithreaded fractal viewer, and a coroutine-based LAN chat.
- **Personal portfolio** — dxcHEL.github.io, built with React and TypeScript.
- **Arch Linux daily-driver setup** — Self-customized system used to build deep, practical knowledge of Linux and its file system.

Courses & Certifications

Advanced C++ Programming and Modern Practices — Microsoft	Feb 2026 – May 2026
Linux Essentials — Linux Professional Institute	Jan 2020 – Apr 2020
Advanced Python — Universidad Marista	Jun 2024 – Nov 2024
Time Management Fundamentals	Feb 2026 – Mar 2026
Foundations of Agile Project Management — Google	Feb 2026 – Mar 2026
Learning How to Learn — Oakland University	Jan 2025 – Jun 2025
Mobile App Development — Google Activate	Nov 2019 – Feb 2020
Intel x86 Processor & Platform Architecture — MindShare	Jan 2022 – Jun 2022

Extra-Curricular Activities

Mexican Mathematics Olympiad (OMM), Mexico City — Bronze award	Mar 2013 – Nov 2014
Mexican Informatics Olympiad (COMI), Mexico City — Participation award	Jan – May 2014
Robot War Staff, IEEE University Branch, Mexico City — Planning and organization staff	Jul 2018 – Jul 2020
Scout, Mexican Scout Association, Morelia — Participation in altruism and teamwork-oriented activities	Jan 2014 – Aug 2015
Emergency Response Helper, ERUM, Mexico — Provided relief support following the 2017 Mexico City earthquake	Sep – Oct 2017